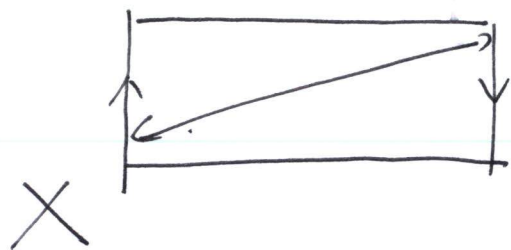


回忆:



$X/\sim$

("~": $p \sim q \Leftrightarrow p$ 与 q 粘到同一点)

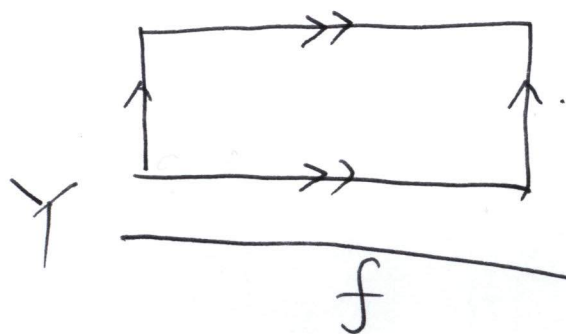
$S^1 \downarrow \varphi$



重新叙述:

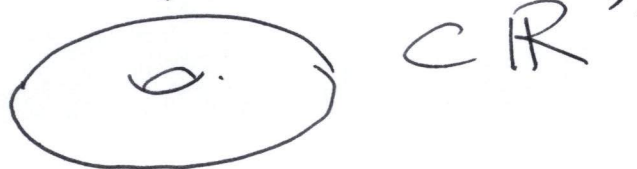
$X/\sim$ 可嵌入到 $\mathbb{R}^3$ 中.

$Y/\sim$ 可嵌入到 $\mathbb{R}^3$ 中.



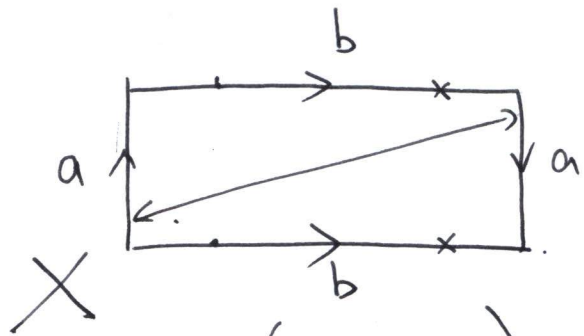
$Y/\sim$

$S^1 \downarrow \varphi$



定义 (嵌入 (embedding)), X, Y : top sp, $\varphi: X \rightarrow Y$, φ 称为一个嵌入, if: $\varphi: X \rightarrow \varphi(X)$ 为一个同胚.
子空间拓扑

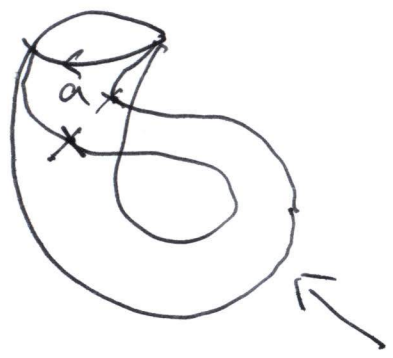
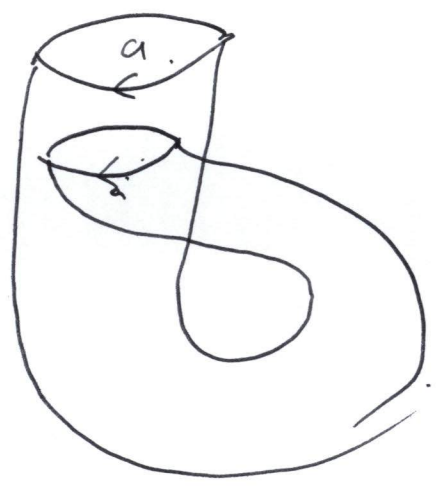
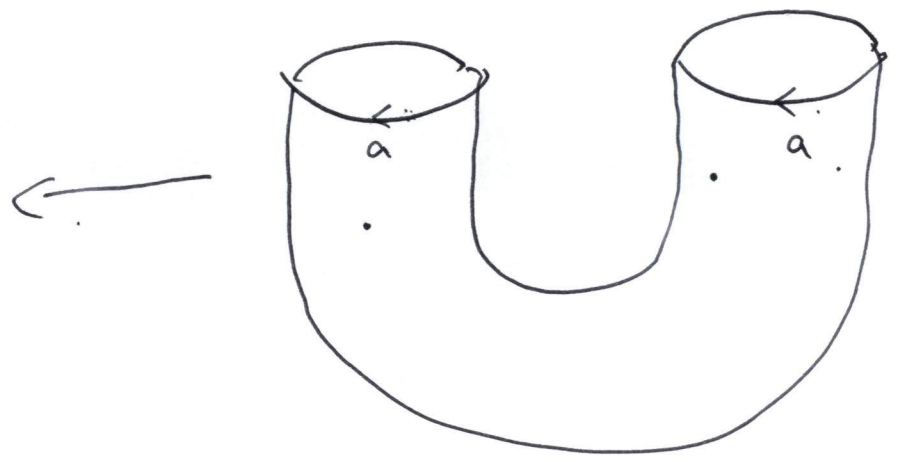
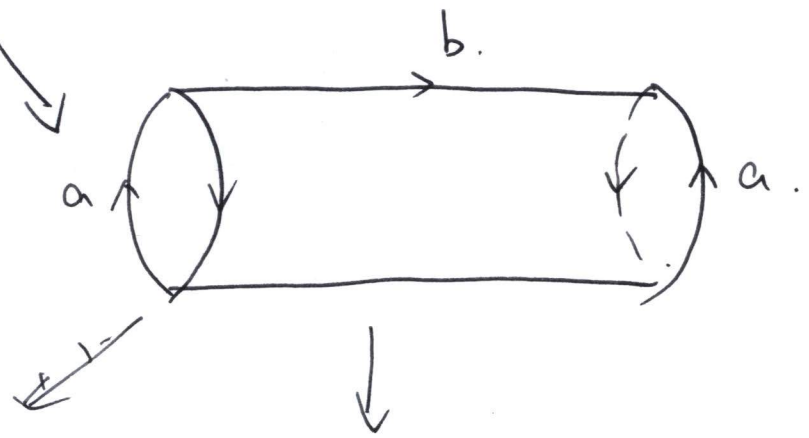
例子 (Klein 瓶)



粘合方式 $\leadsto$ " $\sim$ "

$X/\sim$

Klein 瓶 $:= X/\sim$



事实: Klein 瓶无法嵌入到 $\mathbb{R}^3$ 中.

[c.f. Hatcher. Alexander duality]

Armstrong 习题: Klein 瓶可嵌入到 $\mathbb{R}^4$ 中

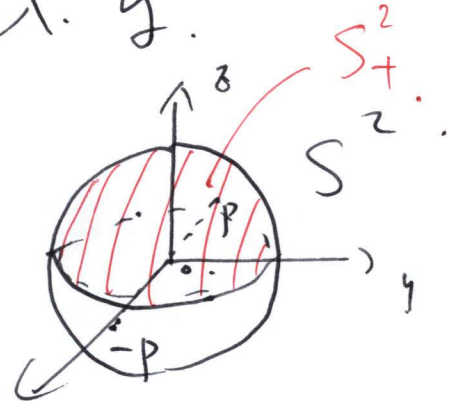
例 (实射影空间).

$$\mathbb{RP}^n := (\mathbb{R}^{n+1} - \{(0, \dots, 0)\}) / \sim.$$

$$\sim: 'x \sim y' \Leftrightarrow \exists \lambda \in \mathbb{R}^* \text{ s.t. } x = \lambda \cdot y.$$

$\mathbb{RP}^n \cong$ “ S^n 沿对径点粘所得空间”
按

$$\pi: \mathbb{R}^{n+1} - \{(0, \dots, 0)\} \longrightarrow \mathbb{RP}^n \text{ 商映射.}$$



$$\begin{array}{ccc} \bigcup_{S^n} & \pi|_{S^n}: S^n \longrightarrow \mathbb{RP}^n & \text{连续满射} \\ \Rightarrow \pi|_{S^n} \text{ 为粘合映射.} & \uparrow & \uparrow \\ & \text{紧} & \text{Hausdorff 空间.} \end{array}$$

$$\Rightarrow \mathbb{R}P^n \cong S^n / \sim_{\pi|_{S^n}}$$

$$'\sim_{\pi|_{S^n}}': \forall p, q \in S^n, p \sim_{\pi|_{S^n}} q, (\Leftrightarrow) \pi(p) = \pi(q)$$

$$(\Leftrightarrow) p = q \text{ or } p = -q$$

$\mathbb{R}P^n \cong B^n$ 按 ^{边界的} 对径点粘所得空间.

$$\pi|_{S_+^n}: S_+^n \rightarrow \mathbb{R}P^n \text{ 连续满射.}$$

$\uparrow$ 紧 $\uparrow$ Hausdorff

$$S_+^n = \left\{ (x^1, \dots, x^{n+1}) \mid \begin{array}{l} (x^1, \dots, x^{n+1}) \\ \in S^n \\ x^{n+1} \geq 0 \end{array} \right\}$$

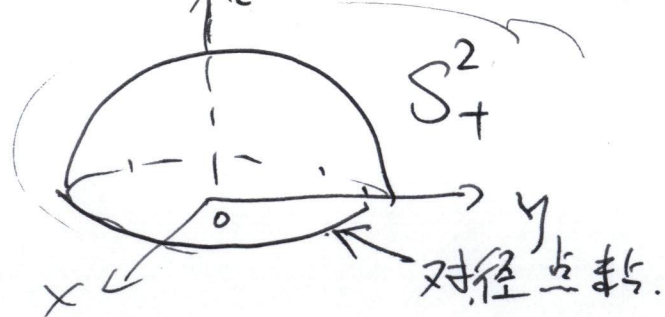
B^n

$\Rightarrow \pi|_{S_+^n}$ 为粘合映射.

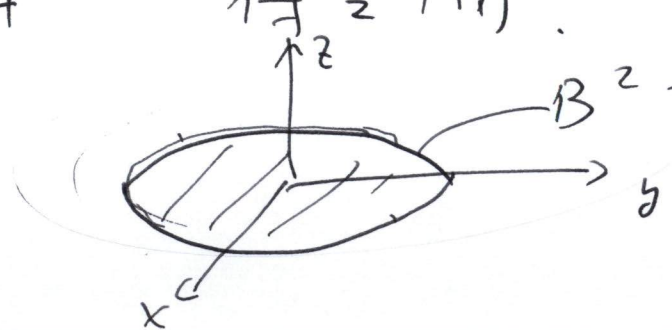
边界按

$$\Rightarrow \mathbb{R}P^n \cong S_+^n / \sim_{\pi|_{S_+^n}} = S_+^n \text{ 的对径点粘合所得空间.}$$

$n=2$.



$\cong$



$\Rightarrow \mathbb{R}P^2$
 $n=2$ 的
 情形.

Lemma. 设 X, Y top. $\varphi: X \rightarrow Y$ 同胚

$\sim: X$ 上等价关系. 通过 φ , " $\sim$ " 定义了 Y 上
等价关系 " $\sim'$ ". ($\forall y_1, y_2 \in Y, y_1 \sim' y_2 \Leftrightarrow \varphi^{-1}(y_1) \sim \varphi^{-1}(y_2)$),

$$\text{则 } X/\sim \cong Y/\sim'$$

Pf. 留作练习.

$n=2, \mathbb{R}P^2$.

可证明: ① $\mathbb{R}P^2$ 不可嵌入到 $\mathbb{R}^3$.

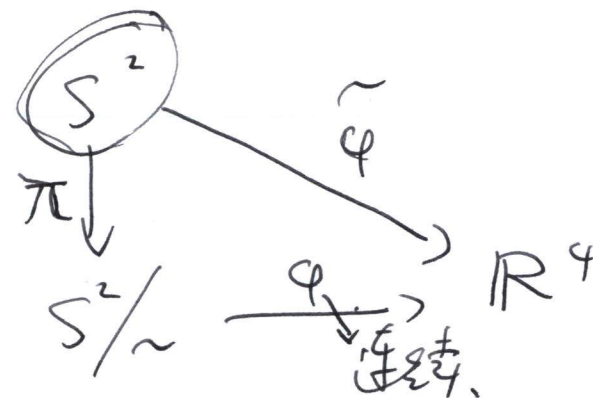
: ② $\mathbb{R}P^2$ 可嵌入到 $\mathbb{R}^4$.

定义: $\mathbb{R}P^2 \longrightarrow \mathbb{R}^4$ 如下.

$$\begin{array}{c} \parallel \\ S^2 \\ \hline \text{对径点相黏} \end{array}$$

定义: $\hat{\varphi}: S^2 \longrightarrow \mathbb{R}^4$
 $(x, y, z) \longmapsto (x^2 - y^2, xy, xz, yz)$

$(-x, -y, -z) \longmapsto (x^2 - y^2, xy, xz, yz)$



$\leadsto \varphi: S^2/\sim \longrightarrow \mathbb{R}^4$
 $\mathbb{R}P^2$ 紧 Hausdorff.

要证 φ 为嵌入, 只需证: $\boxed{\varphi \text{ 为单射}}$ $\rightarrow$ 留作习题.

Lemma: 设 X, Y : top spaces, X 紧, Y : Hausdorff.
 $\varphi: X \rightarrow Y$ 连续单射, 则 φ 为嵌入 ($\varphi: X \xrightarrow{\cong} \varphi(X)$)

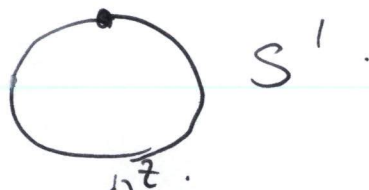
Pf. 只需证: $\varphi^{-1}: \varphi(X) \rightarrow X$ 为连续映射.
 φ^{-1} 连续. $\Rightarrow \varphi(F)$ 紧. $\nearrow$ 要证 $(\varphi^{-1})^{-1}(F)$ 为 $\varphi(X)$ 中闭集 $\varphi(F)$.

#

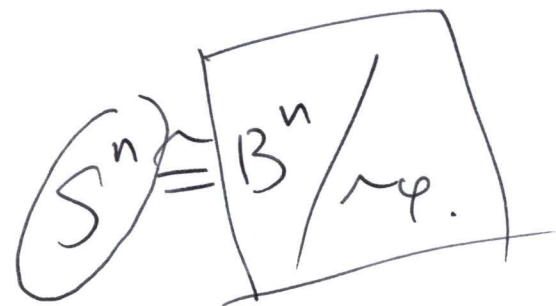
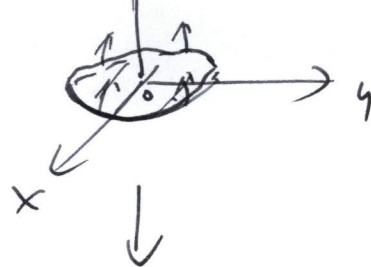
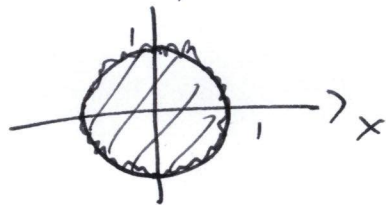
例. $B^n = \{(x^1, \dots, x^n) \in \mathbb{R}^n \mid (x^1)^2 + \dots + (x^n)^2 \leq 1\}$.

将 B^n 的边界粘成一个点.

$n=1$.



$n=2$



$$x \sim y \iff \varphi(x) = \varphi(y)$$

猜: S^n .

只需构造连续映射:

$\varphi: B^n \rightarrow S^n$ 满足:

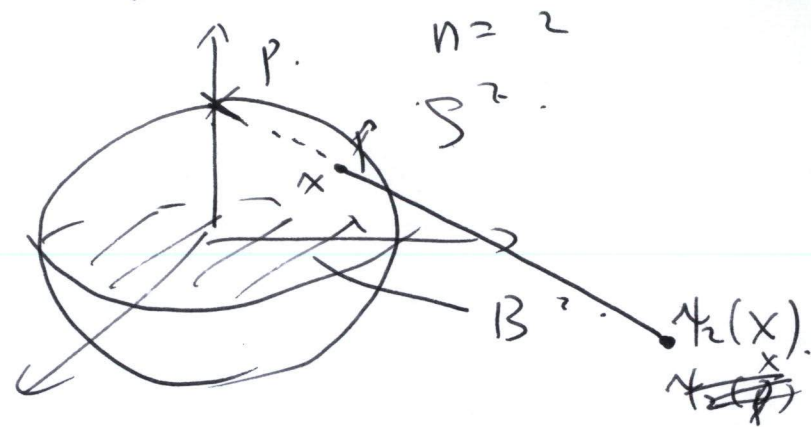
- ① $\varphi|_{(B^n)^\circ}$ 单射.
- ② $\varphi(\partial B^n) = \{p\}$

$$\gamma_1: (B^n)^\circ \xrightarrow{\cong} \mathbb{R}^n. \quad x \mapsto \frac{x}{1 - \|x\|}$$

$$\gamma_2: S^n \setminus \{p\} \xrightarrow{\cong} \mathbb{R}^n$$

$$\underline{\gamma_2^{-1} \circ \gamma_1}: (B^n)^\circ \rightarrow S^n \setminus \{p\}$$

同胚.

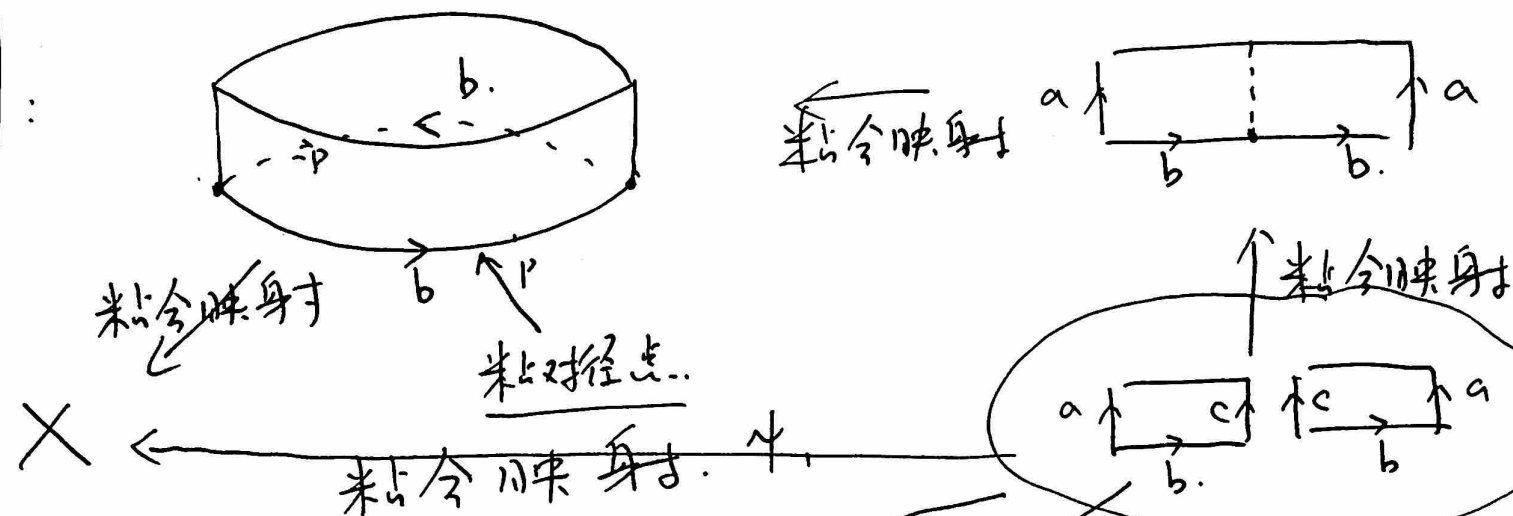


$$\text{定义 } \varphi: B^n \rightarrow S^n$$

$$x \mapsto \varphi(x) = \begin{cases} \gamma_2^{-1} \circ \gamma_1, & x \in (B^n)^\circ \\ p, & x \in \partial B^n. \end{cases}$$

验证 φ 为连续映射 2) 题.

例:



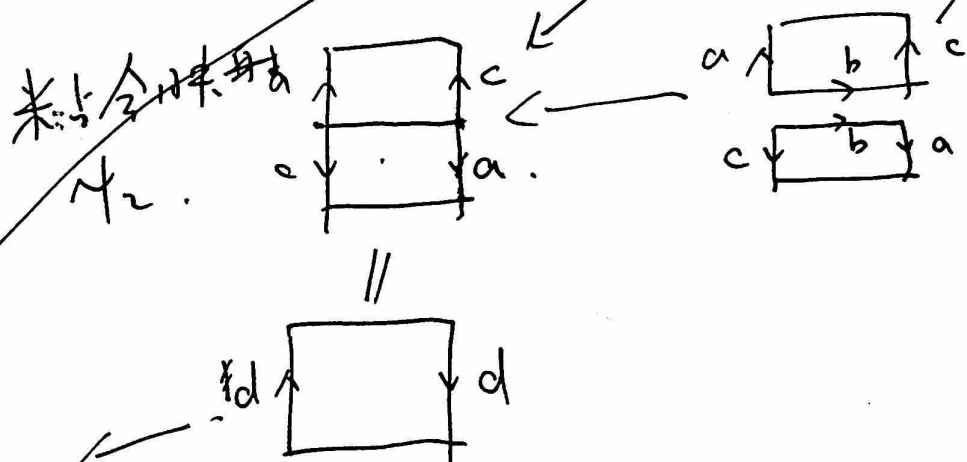
$$X \cong M.$$

$$Y \xrightarrow{\gamma_1} X$$

$$\xrightarrow{\gamma_2} M.$$

$$X \cong Y / \sim_{\gamma_1}$$

$$M \cong Y / \sim_{\gamma_2}$$



$$\sim_{\gamma_1} = \sim_{\gamma_2}.$$

Lemma: 粘合映射之复合仍为粘合映射 (问题!)



下面：把一个空间粘到另一个空间上去。

X, Y top spaces, $A \subset X$.
 $f: A \rightarrow Y$ 连续.

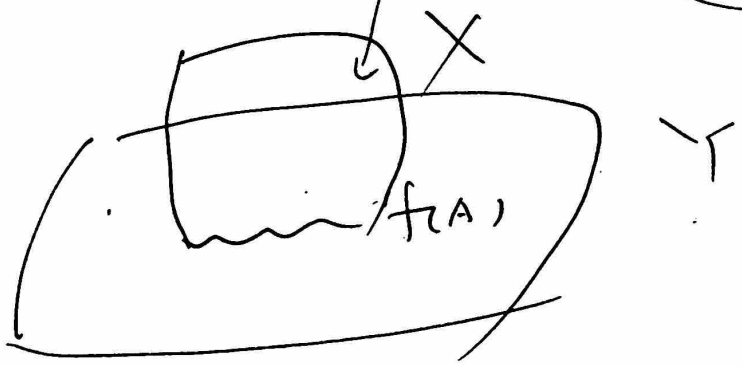


x 与 $f(x)$ 粘.

$f(x') = f(x) \Rightarrow x' \sim x$,
 $f(x)$ 粘成一个点.

$\forall x \in A, f^{-1}(f(x))$ 粘成一个点.
 $(\Rightarrow f(x) \sim x)$

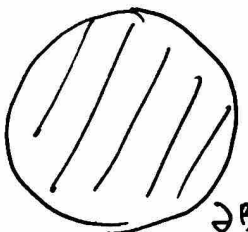
记为 $X \sim_f Y$



拓扑空间
 定义 $X \sim_f Y$ 如下:

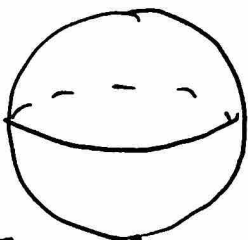
$X \sqcup Y / \sim$
 $\sim$: 简言之, x 与 x' 等价 $\Leftrightarrow x$ 与 x' 粘到同一一点.
 $\sim: \forall x \in A, x \sim f(x)$
 所有点与自身等价.
 $(X \setminus A) \sqcup (Y \setminus f(A))$ 只与自身等价.

例: $X = B^2$



$\partial B^2 = S^1$

$Y: p \rightsquigarrow$



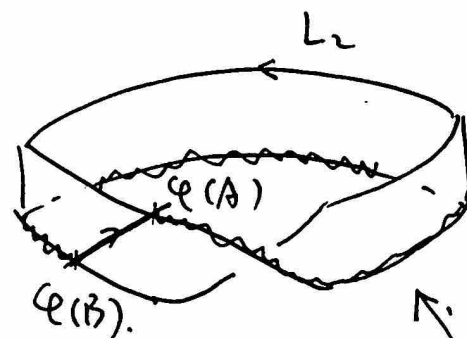
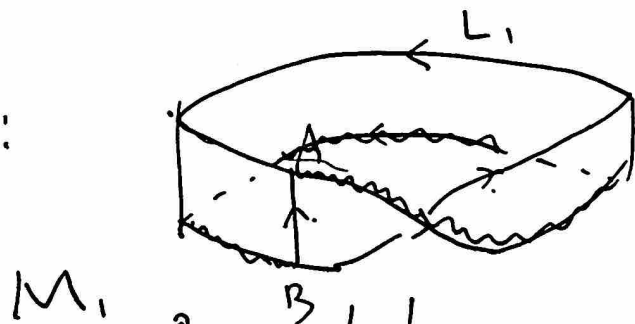
S^2

$f: \partial B^2 \rightarrow Y \rightsquigarrow x \mapsto p$

$X \sim_f Y$

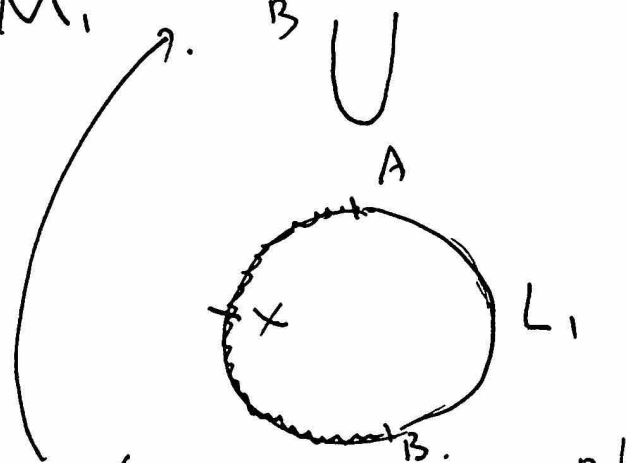
$X \perp Y$

例:



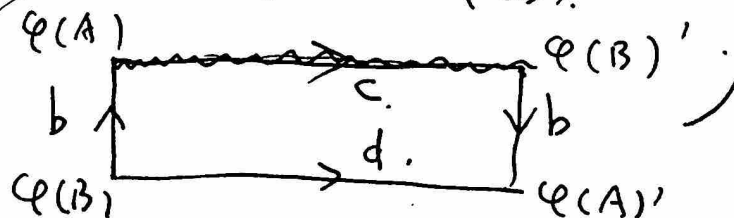
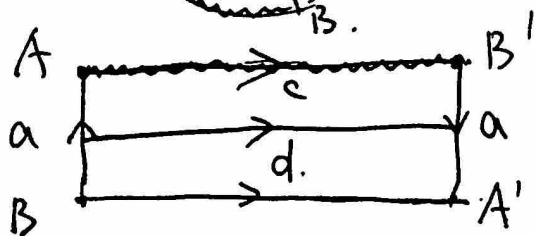
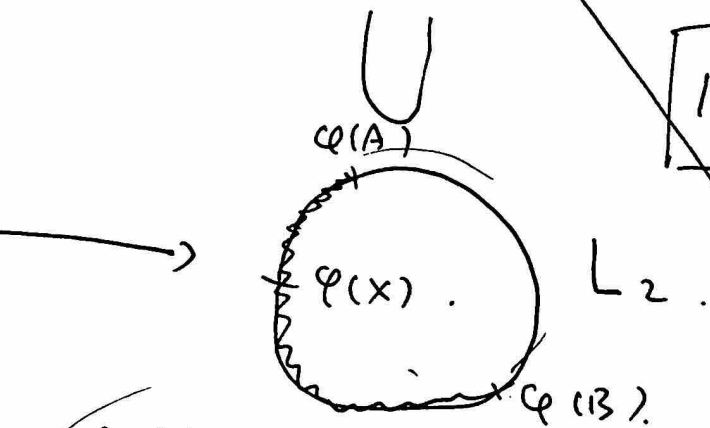
M_2

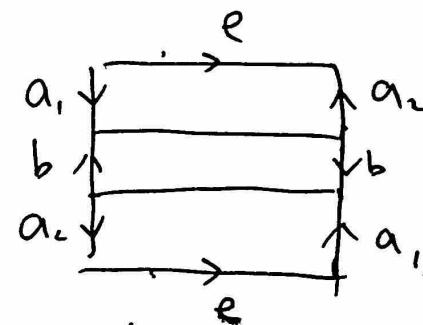
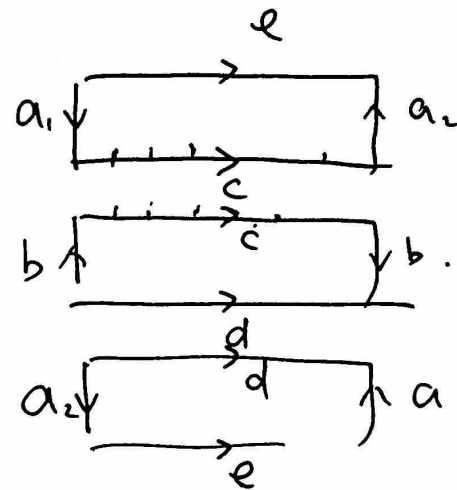
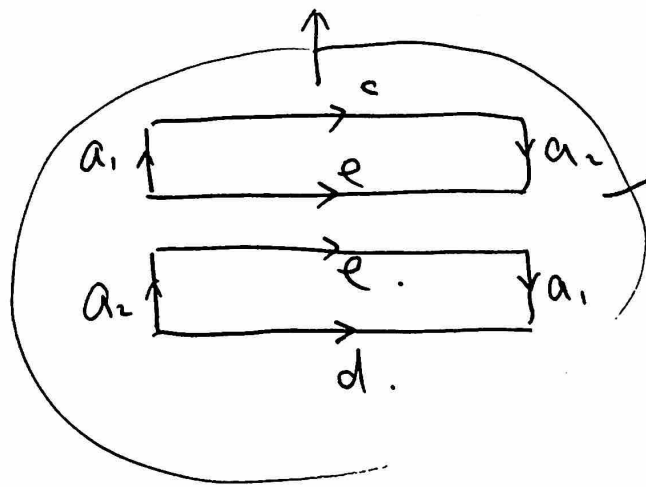
$M_1 \sim_\varphi M_2$



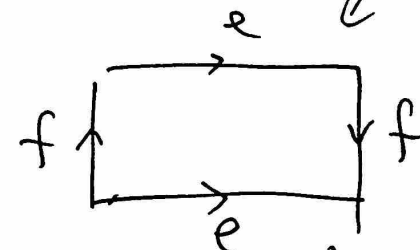
φ

$\cong$



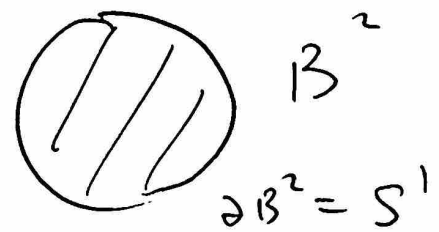


Klein 瓶.



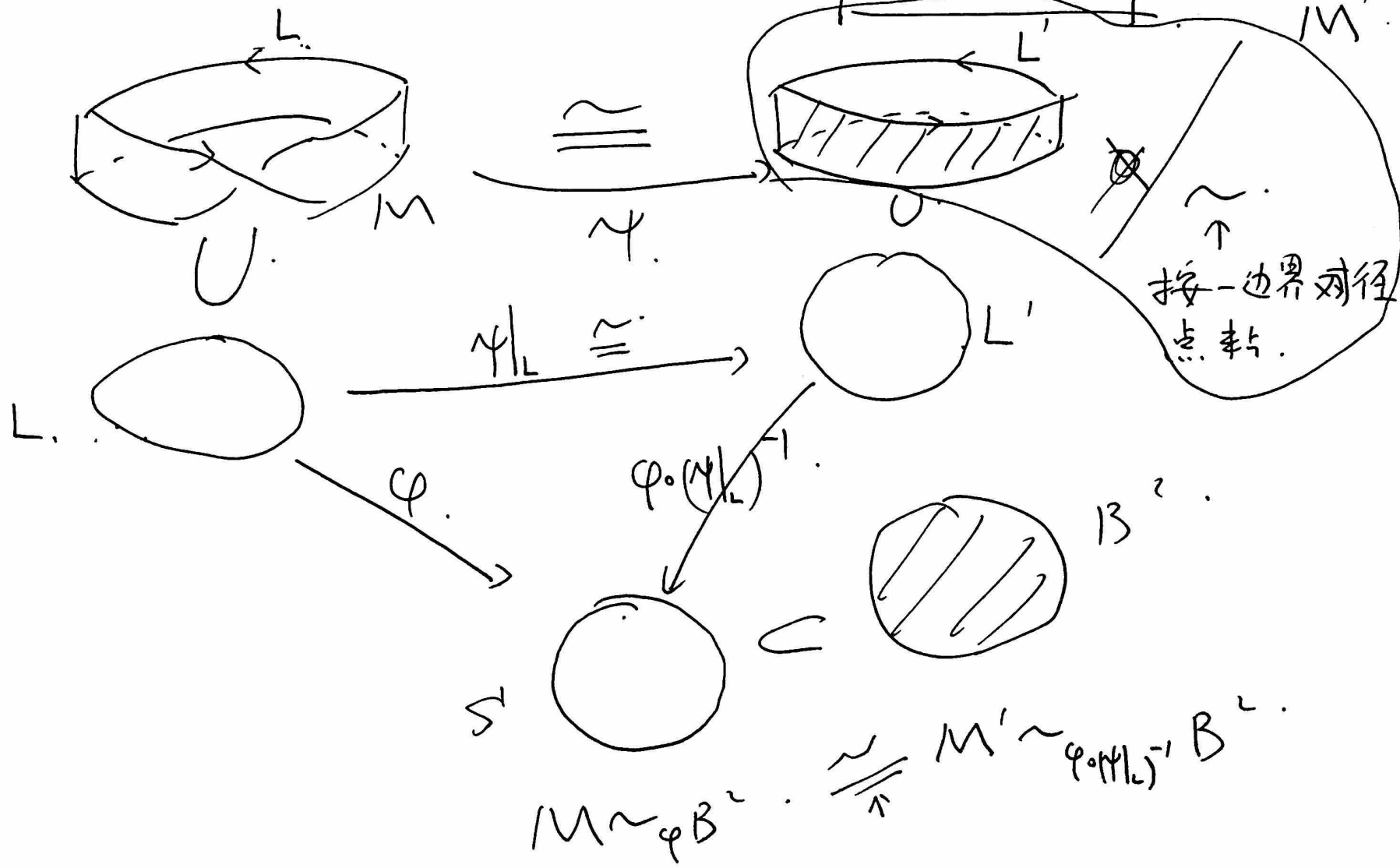
10) 题: $\varphi: L_1 \xrightarrow{\cong} L_2$ 选取不同, 所得空间是否相同?

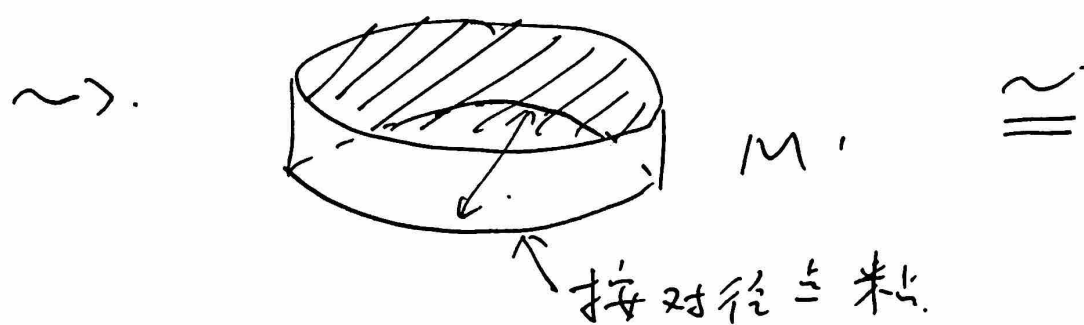
例1:



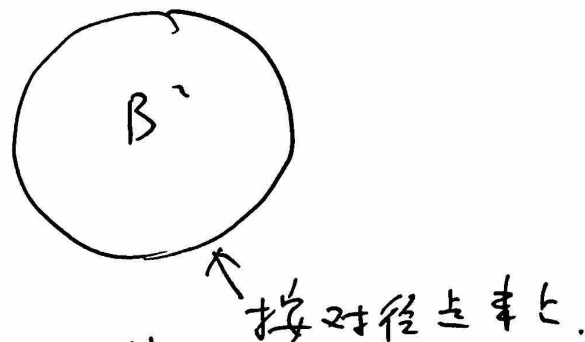
选取 $\varphi: \partial M \xrightarrow{\cong} \partial B^2$.

已证:





$\cong$



$\cong$

$\mathbb{R}P^2$

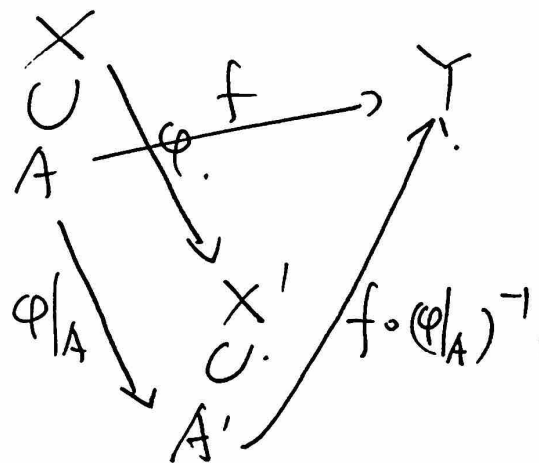
Lemma. 设 X, Y top. spaces, $A \subset X$,

$f: A \rightarrow Y$ 连续, 设 X' : top space,

$\varphi: X \rightarrow X'$ 同胚, $\varphi(A) = A' \subset X'$

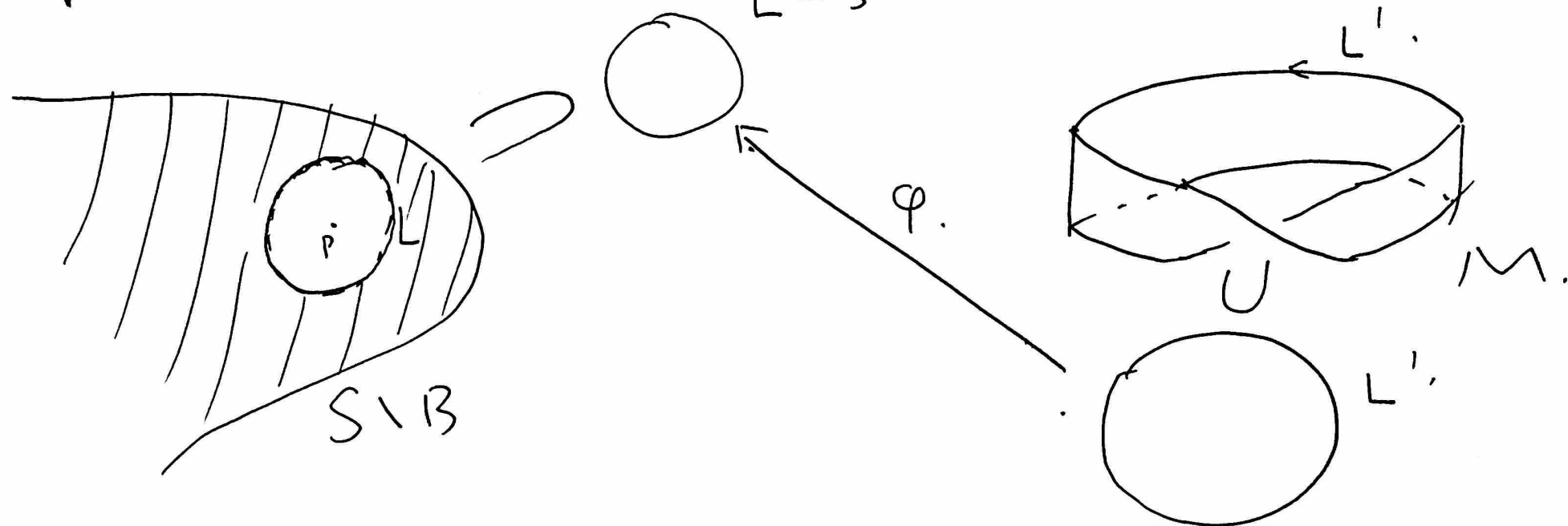
$$X \sim_f Y \cong X' \sim_{f \circ (\varphi|_A)^{-1}} Y$$

(留作习题)



Rmk. 设 S 为一个曲面.

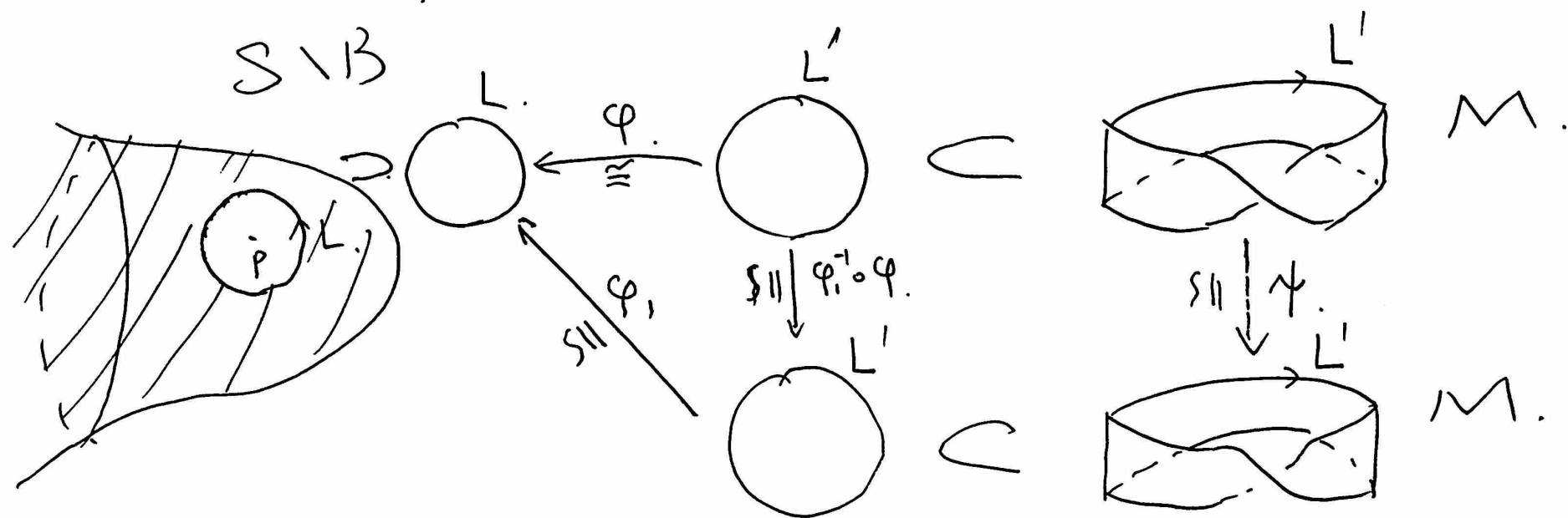
$\forall p \in S$, 以 p 为圆心挖掉一个圆盘 B ,
 $L \cong S'$



问: 同胚
 φ 之选取不同, 是否影响粘合的结果.
 i.e. 若 $\varphi' : L' \xrightarrow{\cong} L$ 为另一同胚, 问:

$$M \sim_{\varphi} (S \setminus B) \cong M \sim_{\varphi'} (S \setminus B)$$

S 曲面, $p \in S$, $B \subset S$ 包含 p 的一个开圆盘.



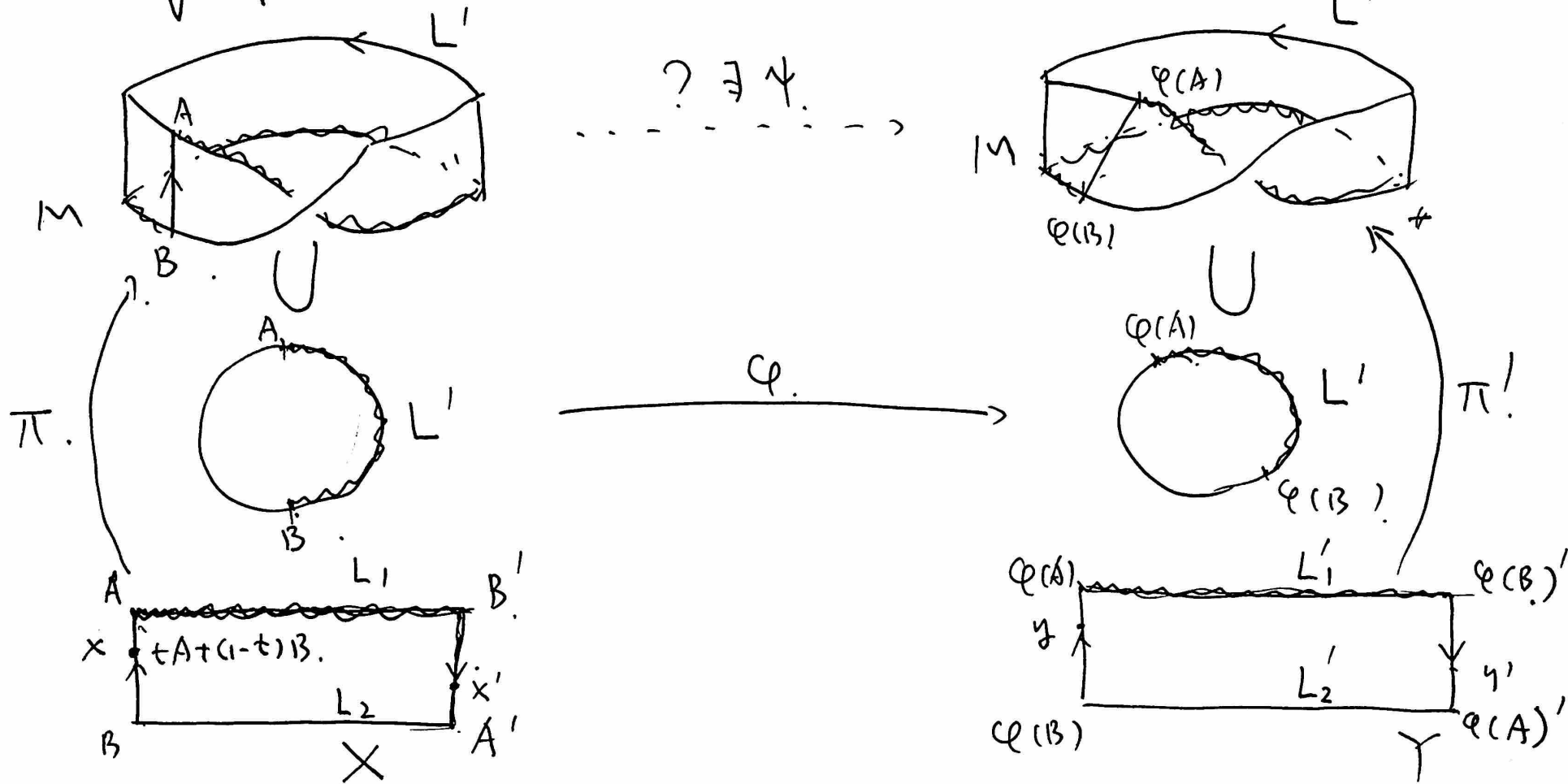
$$M \sim_{\varphi}(S \setminus B) \stackrel{?}{\cong} M \sim_{\varphi_1}(S \setminus B).$$

若 $\exists \psi: M \xrightarrow{\cong} M$, s.t. $\psi|_{L'} = \varphi_1^{-1} \circ \varphi$. 则

$$M \sim_{\varphi}(S \setminus B) \cong M \sim_{\varphi_1}(S \setminus B).$$

问题归结于: 给定 ψ 从 $\partial M \rightarrow \partial M$ 的同胚, 是否可将之
延拓到从 M 到 M 的同胚.

$$\forall \varphi : \partial M \rightarrow \partial M.$$



$$\varphi \text{ 诱导 } \hat{\varphi} : L_1 \amalg L_2 \rightarrow L'_1 \amalg L'_2.$$

$$\begin{array}{ll} A \longmapsto \varphi(A), & A' \longmapsto \varphi(A)' \\ B \longmapsto \varphi(B), & B' \longmapsto \varphi(B)'. \end{array}$$

$$\leadsto \hat{\varphi} \text{ 可 } \text{延拓到} \partial X \rightarrow \partial Y \quad tA + (1-t)B \mapsto t\hat{\varphi}(A) + (1-t)\hat{\varphi}(B).$$

进一步 $\tilde{\varphi}$ 可延拓到:

$$\tilde{\varphi}: X \xrightarrow{\cong} Y \quad \text{且保持对径点.}$$

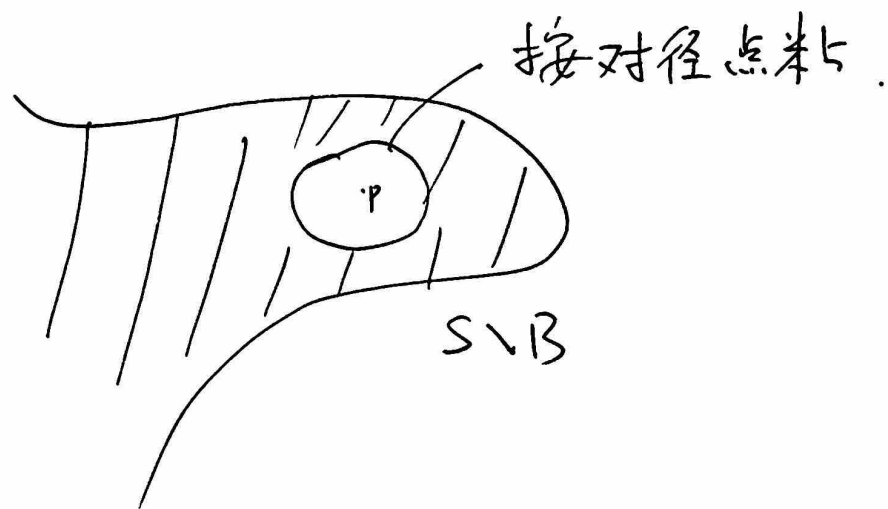
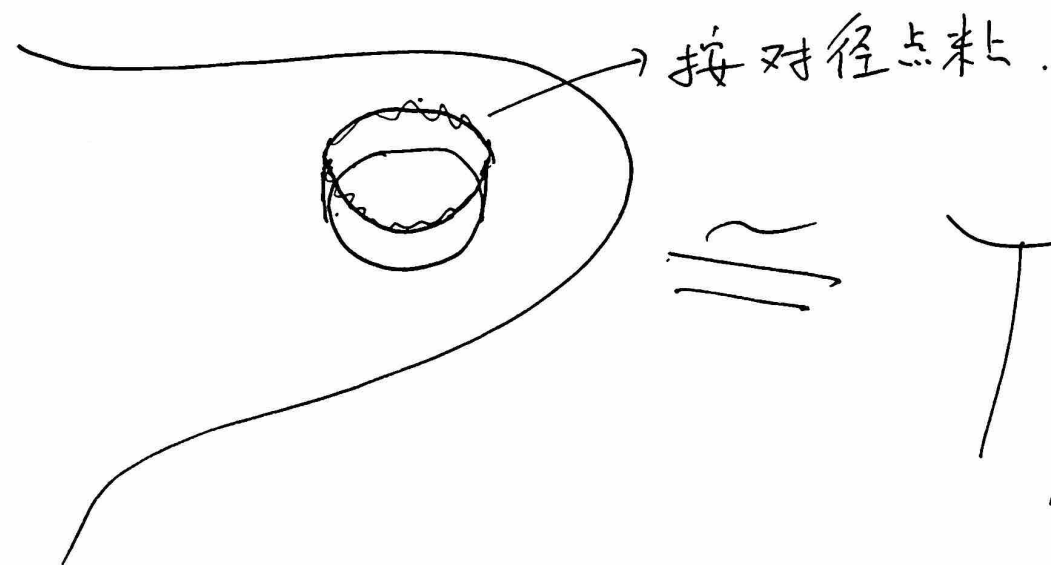
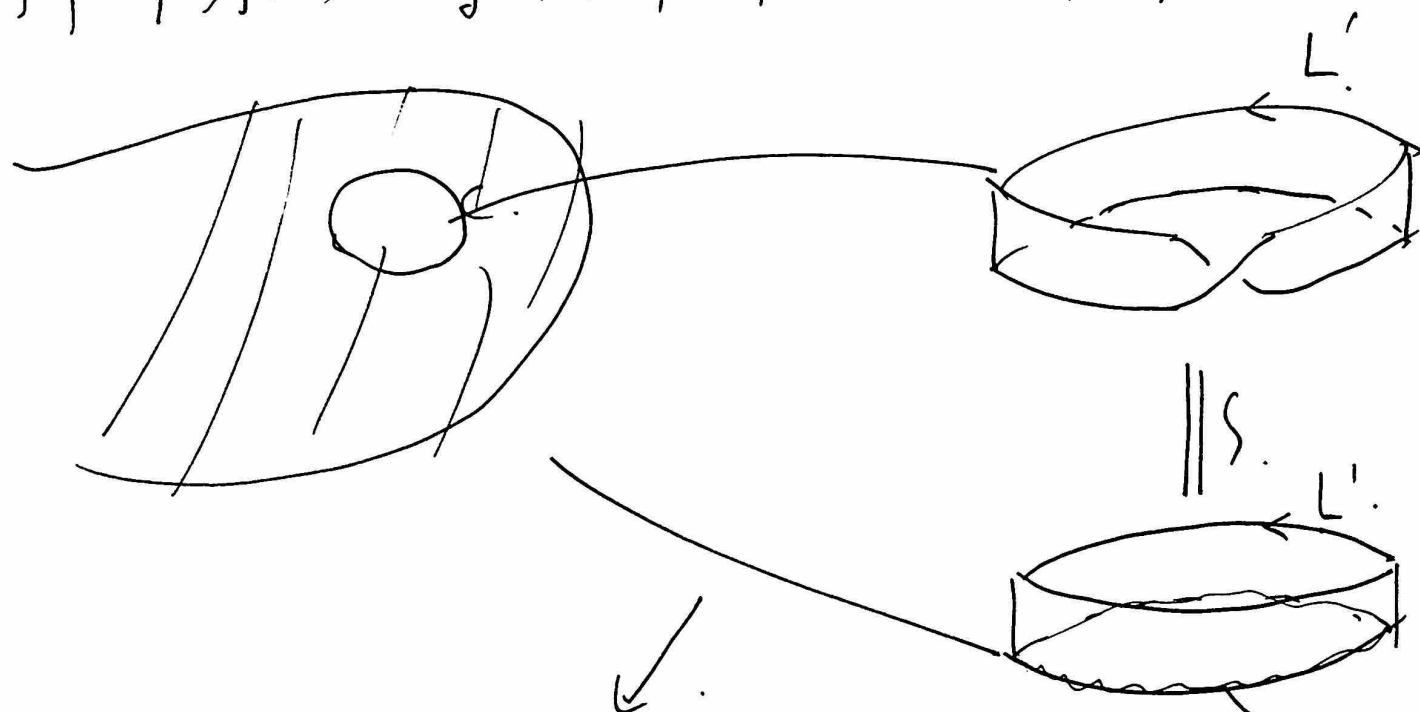
Lemma: B^2 : 单位闭圆盘, $\varphi: \partial B^2 \rightarrow \partial B^2$ 同胚, 则 φ
可延拓为 $B^2 \xrightarrow{\cong} B^2$ ┐

$\tilde{\varphi}$ 决定了一个同胚 $\psi: M \rightarrow M$.

$$\begin{array}{ccc} X & \xrightarrow{\tilde{\varphi}} & Y \\ \pi \downarrow & & \downarrow \pi' \\ M & \xrightarrow[\cong]{\psi} & M \end{array}$$

总结: 将 Mobius 带的边界粘合 $S \setminus B$ 这种手术, 不依赖于
边界之间的同胚的选取.

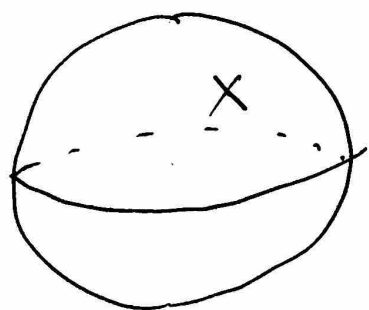
此种手术又等价于下面的手术：



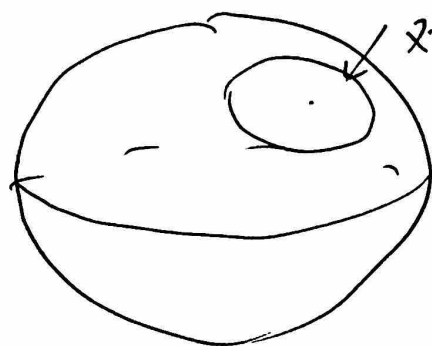
$S \setminus B$: { ① 将 M 沿边界粘到 $S \setminus B$ 上.
② 将 $S \setminus B$ 的边界 (∂S), 按对径点粘.

2. 在 S 上安一个 crosscap.

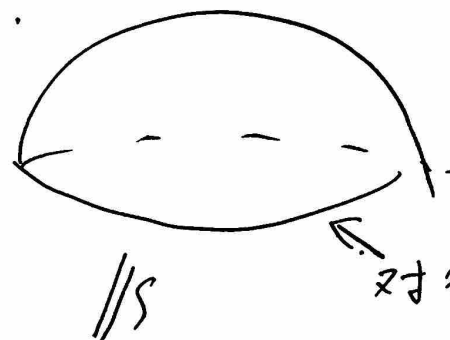
将证明: 在球面上安 crosscap 所得拓扑空间, 仅与所安 crosscap 的个数有关.



$M(1)$.

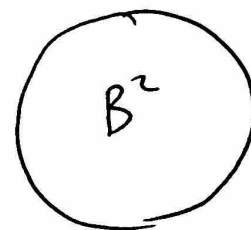


$\cong$

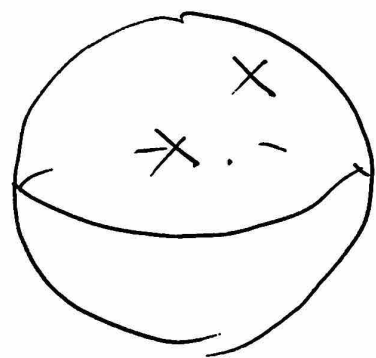


$//S$

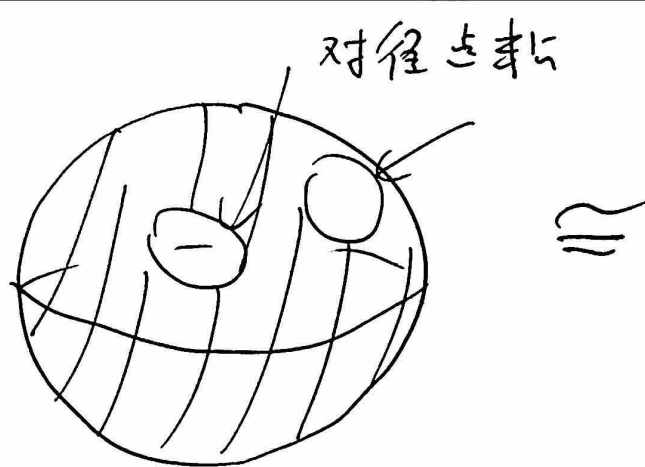
$\mathbb{RP}^2 \cong$



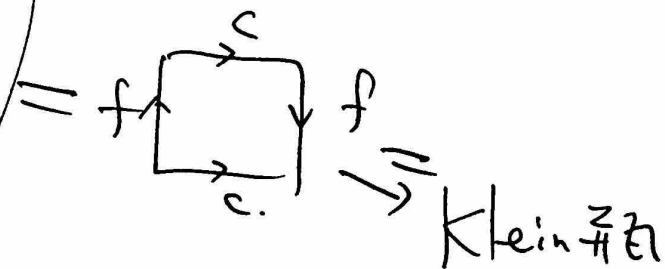
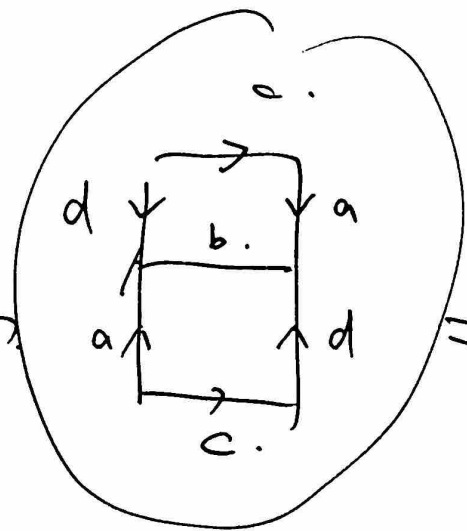
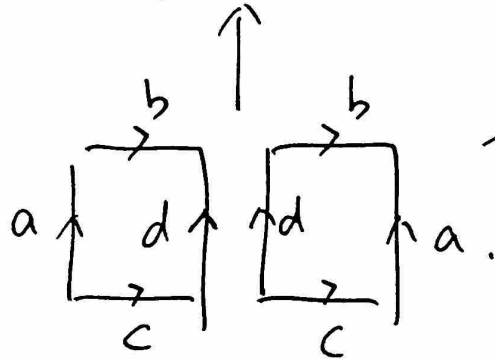
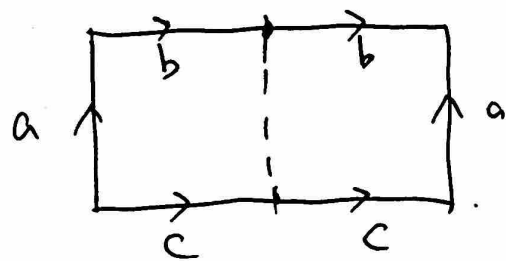
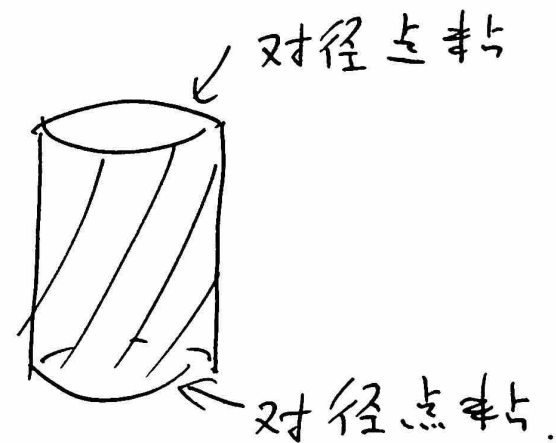
$\leftarrow$ 对径点粘.



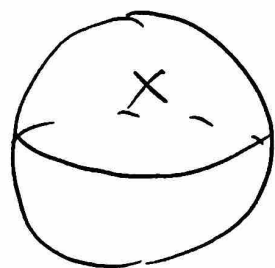
$M(2)$.



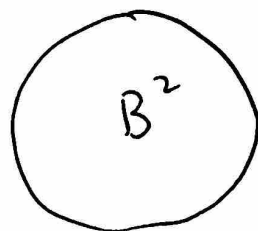
$\cong$



$M(1):$

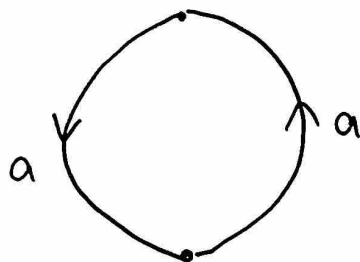


$\cong$

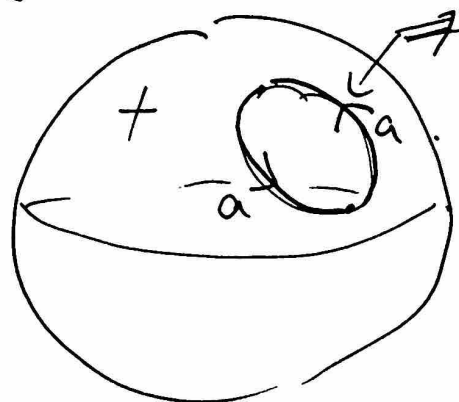
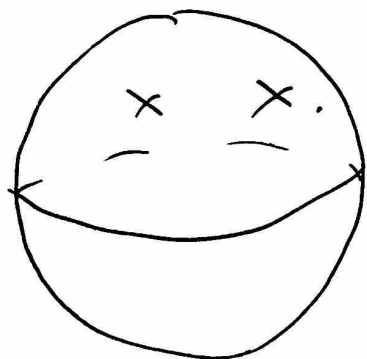


对径点粘.

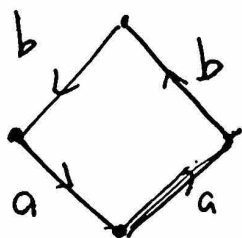
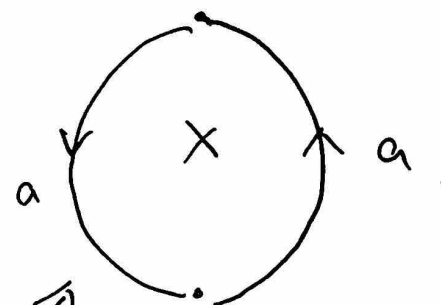
2边开}



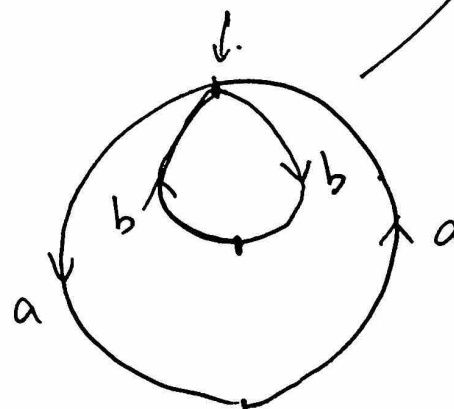
$M(2):$



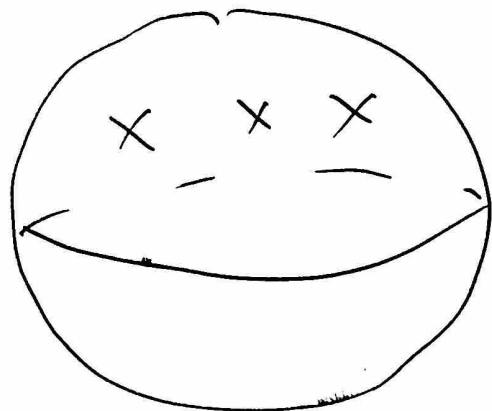
$\cong$



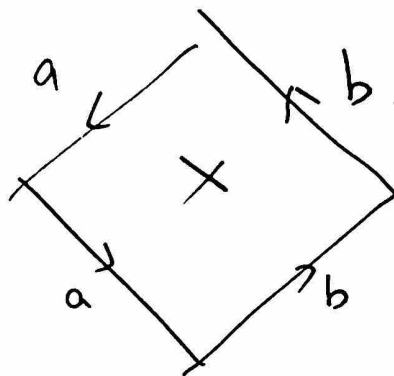
4边开}



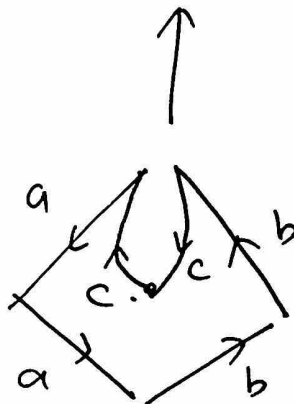
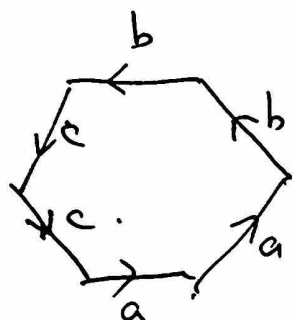
$M(3)$:



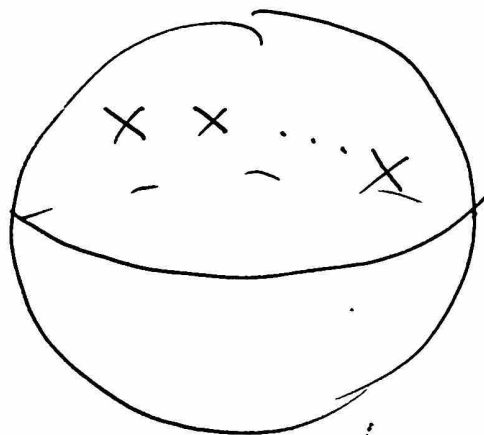
$\cong$



2.3 边开}

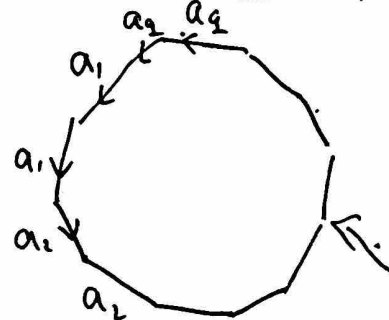


$M(g)$:



$\cong$ ←

2g 边开}



之称为g的
不可定向曲
面。

$M(g) :=$ 在球面上安g个 ~~crosscap~~ crosscap 所得空间